Is there a mass above which quantum superposition stops? Registered readings of mass and distance, the 2026 interference record and a prediction dated to 2035

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Abstract

Standard quantum mechanics has no mass at which a superposition stops: large objects look classical because of decoherence, the environment that keeps recording where they are. Collapse models (Ghirardi–Rimini–Weber, CSL, Diósi–Penrose) add such a limit by hand. The framework says that a threshold exists: below a certain mass level the principal GCD of the point of observation and the GCD of the diffraction that generates the object become equal, and the object spreads over several positions (see the box above). The box gives no number. This deposit tries to give the threshold a number, or to say clearly why it cannot. A second part tests two later remarks of the author: the limit cannot be fixed as a mass, because a calculation of the «fractional distance» is missing; and large distant bodies stay fixed because their size makes up for the distance. It was announced in deposit 2.64 (DOI 10.5281/zenodo.23610071).

Ugo Lanciano, October 2026. Licence CC BY 4.0 (texts); code MIT. The same text in Italian is in LEGGIMI_IT.md.

Deposit 2.64 (mass as a contained rhythm) left one question aside: the mass threshold of the quantum world, «treated in a separate work with a prediction dated to 2035». This is that work. It can be read on its own. It has two parts, done with the same method: the threshold as a mass (22 October 2026) and the role of distance (25 October 2026, §§3.3, 4.6, 5.5), added before upload after two remarks of the author.

In brief

Small things behave in a strange way. An atom, or a molecule, sent through a set of slits does not go through one slit or the other: it goes through all of them at once, and on the screen it leaves the stripes of a wave. Physicists call this a **superposition**: the object is spread over several positions at the same time. Large things are never seen like this: a chair is always in one place. Where does the change happen, and why?

Standard quantum mechanics has no mass at which the change happens. It explains the classical look of large objects with **decoherence**: the environment (air molecules, light, heat) constantly «looks» at a large object, and the stripes are washed out before anyone can see them. Some physicists think there is more: a true limit, present even in a perfect vacuum, beyond which a superposition cannot survive. These are the **collapse models**; the best known ties the limit to gravity (Diósi and Penrose).

The author's framework says that such a limit exists. In its summary published on every record («MM in brief»), when an object falls below a certain mass level, two quantities of the framework, the «principal GCD» of the point of observation and the «GCD of the diffraction» that generates the object, become equal, and the object spreads over several positions: we observe it as probability. That summary gives no number for the threshold. This work tries to give it one, or to say clearly why it cannot. After its first part, the author added two remarks: the limit cannot be fixed as a mass, because what is missing is the calculation of a «fractional distance»; and at an adequate distance even a large object could take on quantum features, while large distant bodies stay fixed because their size makes up for the distance. The second part tests these remarks.

We did three things, in each part.

1. **The data.** From 20 papers downloaded from arXiv we listed what experiments have seen: nine experiments with quantum behaviour, from single atoms to a vibrating crystal of 16 micrograms, and the limits on collapse models. For the second part, 22 more sources: decoherence measured against the separation and the number of scattered photons, quantum correlations over distances from 20 cm to 1 203 km, Wigner's-friend experiments,

stellar diameters measured by interferometry, and the threshold of the naked eye.
2. **Readings.** We translated the thesis into precise criteria: ten for the mass, eight for the distance. Each says which physical quantities stand for the terms of the framework and where, as a consequence, the threshold lies. We registered all of them, with the rules for the outcome, before doing any calculation. Each translation is a choice made by the AI, not the author's words.
3. **Tests.** For the readings that survive, a prediction dated to 31 December 2035 with rules that anyone can apply, and, for the distance, a test with data that already exist, registered before the sources that decide it were read.

The main results are five.

- **For distance, the limit is not a mass alone.** In the two readings of the fractional distance that hold, it is a ratio between the size that has to be told apart and the «cell» of the copies that do the telling, together with the number of copies. (The readings of the first part with a mass threshold at the Planck scale remain possible, untested.) The two readings have a similar form; that they are one rule of the framework is a proposal. For the state of an object, the copies are the photons and molecules the environment carries away, and the cell is the scale at which one copy tells two positions apart (for light, about half a wavelength): below it each copy takes away only a small part of the coherence, and a crystal of 16 micrograms stays quantum over $2\cdot10^{-18}$ m because about 10^{34} thermal copies would be needed to fix it. For the way a distant observer reads an object, the cell is the diffraction cell of the observer's aperture at the distance of the object. Two stars of almost the same mass fall on opposite sides: the disc of Sirius B (1.02 solar masses, 2.64 parsec) is 20–60 times smaller than the finest cell of today's interferometers (a calculation from its radius), while τ Ceti (0.78 solar masses, farther, 3.65 pc) and 61 Cyg A (0.67–0.74, 3.49 pc) are resolved. With this instrumental classification, the sample does not support a mass-only limit of the reading that grows with distance. The reading «what matters for the state of an object is how far the observer is» does not follow from the data: heating fullerene molecules, with the same apparatus and the same observer, takes their interference from 47 % to 7 % and then to zero, through photons that nobody collects.
- **Three readings do not follow from the data.** The threshold as an internal acceleration compared with the galactic scale a_0 (a reading from the author's book), the threshold at the mass of the proton or of the electron, and a_0 written as a mass. In 2026 clusters of about 7 500 sodium atoms (172 000 atomic mass units), held together by accelerations 26 orders of magnitude above a_0, were seen to interfere; a caesium atom, 132 times the proton mass, kept a superposition of 1.9 micrometres for 70 seconds.
- **Five readings remain possible but cannot be tested today:** two thresholds near the Planck mass (22 micrograms, and 15.4 micrograms in the «two walls» form), a threshold at the mass of the observer (70 kg) and two thresholds of the Diósi–Penrose type. No experiment has yet put the centre of mass of an object heavier than about $3\cdot10^{-22}$ kg into a superposition as large as the object itself; in today's interference experiments the gravitational effect is at most $4.2\cdot10^{-10}$ of what would be needed to see it.
- **The reading with the clock of deposit 2.40 sits exactly on the 2026 record** (ratio 1.24), but it is not a general threshold. It coincides with the design condition of grating interferometers (the Talbot condition); atom interferometers, where the superposition is created by laser pulses, go beyond it by seven orders of magnitude. It describes a condition of the apparatus, that is, of the reading, not a property of the object.
- **The dated prediction is registered** (time stamp 22 October 2026, 13:00:30 UTC; deadline 31 December 2035), with a program that applies its rules (45 tests out of 45). The heaviest planned experiment, the space mission MAQRO in its most optimistic study scenario, would reach 0.095–0.35 of the gravitational effect; within the limits of the published study, at most 0.05. The most likely outcome in 2035 is «not assessable».

No experiment has met a threshold so far. All the data are explained by standard quantum mechanics with decoherence; the reading «no intrinsic threshold, the threshold is decoherence» is compatible with them. The two rules of the fractional distance that hold coincide with known physics (decoherence, and the wave optics of the light that carries the image): the framework admits them; an experiment can put them to the test, but cannot tell them apart from standard physics.

How to read this text. Section 2 explains the thesis without formulas. Section 1 summarises what physics already knows and what the author's framework had already said: on a first reading it can be skimmed. Section 3 says how the thesis was tested, §4 gives the results, §5 the dated prediction, §6 what is not claimed. The second part, on distance, is in §1 (last paragraphs), §2.2, §3.3, §4.6 and §5.5. Technical words are explained the first time they appear and again in the glossary. Numbers such as 2.38, 2.40, 2.48 and 2.64 refer to earlier works of the author deposited on Zenodo; the list with DOIs is at the end.

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1. What is already known

Quantum mechanics has no mass threshold. Nothing in the theory stops a superposition when the object grows. What

makes large objects look classical is decoherence (Joos and Zeh 1985; Zurek 2003; Schlosshauer 2019): every collision with a gas molecule or a photon carries away a trace of where the object is, and the stripes disappear. The times are very short for large objects: for a speck of dust spread over a hundredth of a millimetre, decoherence takes about 10^{-31} s in air and 10^{-14} s in the best laboratory vacuum (Schlosshauer 2019, Table 1). For a decoherence threshold the environment decides, not the mass.

Collapse models put a threshold in by hand. Ghirardi, Rimini and Weber (1986) and the continuous version CSL (Pearle 1989; Ghirardi, Pearle and Rimini 1990) add to quantum mechanics a small random process that localises objects: rare for one atom, fast for many atoms together. Two numbers fix it, a rate λ and a length r_C. The values proposed by Ghirardi, Rimini and Weber ($\lambda = 10^{-16}$ s^{-1} at r_C = 10^{-7} m) have not been excluded yet; larger values have been excluded by X-ray emission from germanium detectors and by mechanical experiments (Carlesso and others 2022; the review of Carlesso and Donadi 2025). Diósi (1987, 1989) and Penrose (1996) tie the collapse to gravity: a superposition of two positions of a mass lasts about ħ/E_G, where E_G is the gravitational energy of the difference between the two arrangements of mass. In its version without free parameters this model is excluded by an underground experiment at Gran Sasso (Donadi and others 2021); in the version with a radius R_0 over which the mass of each nucleus is spread, the same kind of measurement requires $R_0 > 2.54 \cdot 10^{-10}$ m (Majorana Demonstrator 2022, as corrected by its erratum). Older still is Károlyházy (1966).

How «quantum» an experiment is. Nimmrichter and Hornberger (2013) proposed a number, the macroscopicity μ: the larger μ, the more the experiment restricts any modification of quantum mechanics of the collapse type.

What experiments have seen (the inventory, file 01_livello0/INVENTARIO_LIVELLO0_v1.md, with page numbers; u is the atomic mass unit, $1.66 \cdot 10^{-27}$ kg):

	experiment	object	mass	separation Δx	time T
E1	Pedalino and others, Nature 649, 866 (2026)	clusters of 5 000–10 000 sodium atoms, about 8 nm across, in a three-grating interferometer	143 000–197 000 u (mean 172 000 u, $2.9 \cdot 10^{-22}$ kg)	grating period 133 nm	6.1 ms between gratings
E2	Fein and others (2019), second hand	molecules of about 2 000 atoms	over 25 000 u	not read	not read
E3	Bild and others, Science 380, 274 (2023)	a vibration mode of a sapphire crystal, prepared as a «cat state»	effective mass 16.2 μg	$2.1 \cdot 10^{-18}$ m	under 84 μs
E4	Delić and others (2020)	silica nanoparticle cooled to its ground state	$2.8 \cdot 10^{-18}$ kg	zero-point motion only	7.6 μs
E5	Rossi and others (2025)	silica nanoparticle	$1.2 \cdot 10^{-18}$ kg	coherence 73 pm	—
E6	Tomassi and others (2026)	silica nanoparticle, expanded thermal state	$2.0 \cdot 10^{-18}$ kg	coherence 1–7 pm	260 μs
E7	Whittle and others (2021)	a vibration mode of the LIGO mirrors	effective mass 10 kg	zero-point motion only	—
E8	Panda and others, Nature Physics 20, 1234 (2024)	one caesium atom held in an optical lattice	133 u	1.9 μm	up to 70 s
E9	Kovachy and others (2015), second hand	rubidium atoms	87 u	54 cm	not read

E1 holds the record of macroscopicity, μ = 15.45, about one order of magnitude above the previous record. Its stripes have a visibility of 0.10 and 0.08 and follow the quantum prediction, not the classical one. A fact of the data, with no reading: above about $3 \cdot 10^{-22}$ kg no experiment has put an object into a superposition as large as the object itself; heavier objects behave quantum mechanically only over distances of 10^{-11} to 10^{-18} m.

What is planned. The space mission MAQRO (Gasbarri and others 2021; launch foreseen for 2034) studies particles of 10^7 to 10^{11} u in free fall for up to 100 s; its feasibility study gives visible stripes up to 40 s and particles up to 400 nm. The authors of E1 announce a next step at a million atomic mass units and, in a vertical interferometer, another factor of a hundred in mass and in coherence time. Carlesso and others (2022) estimate that reaching the rate of Ghirardi, Rimini and Weber needs about 10^7 u, a superposition of 180 nm and 20 s.

Distance, in the experiments (second part; inventory in 09_distanza_frazionale/01_livello0, with page numbers):

- *Separation against the scale of the copies.* In an atom interferometer one scattered photon removes all the contrast when the two paths are about half a wavelength apart, «the resolving power of an ideal Heisenberg microscope», with revivals beyond (Pfau and others 1994, Chapman and others 1995, as reported by Kokorowski and others 2001); with many photons the contrast falls as $\exp(-\kappa^2 d^2/2)$, and at fixed d almost exponentially with the mean number of photons (Kokorowski and others 2001, $\bar{n}$ from 0.9 to 8.2). Fullerenes heated before an interferometer lose their stripes (47 %, 29 %, 7 %, 0 % at 0, 3, 6 and 10.5 W) through thermal photons that fly off into space; the loss follows the wavelength of the photons compared with the separation (Hackermüller and others 2004). With a residual gas the visibility falls as $\exp(-p/p_0)$, in agreement with decoherence theory with no adjustable parameter (Hornberger and others 2003). The copies carried away count even if nobody collects them.
- *Distance between the parts and from the source of the settings.* Correlations beyond the Bell bound between atoms at 398 m ($S = 2.221 \pm 0.033$), electron spins in two diamonds at 1.3 km (2.42 ± 0.20), photons sent from a satellite to stations 1 203 km apart (2.37 ± 0.09), with settings chosen by the light of stars hundreds of light years away and of quasars billions of years old; entanglement between two mechanical oscillators 20 cm apart; a quantum eraser at 55 m (visibility 0.95) and at 144 km (0.75–0.78, over a link attenuated by 33 dB). These are different set-ups, not a controlled series: they show quantum behaviour at every separation tried, and none of them changes only the distance between a system and whoever records it.
- *Wigner's friend.* With the friends' records coded in photonic paths, or in a few qubits of a quantum computer, the observers outside see correlations that the absoluteness of the friends' outcomes (with locality and no superdeterminism) does not allow (Proietti and others 2019, $S = 2.416 \pm 0.075$; Bong and others 2020; Zeng and others 2025; a preprint of 2026). No experiment has a friend with a macroscopic mass.
- *Reading distant objects.* The light of a distant star is coherent across an aperture smaller than about λ/θ (θ the angular diameter; van Cittert–Zernike theorem): interferometers measure stellar diameters from the fringes (α Cen A 8.502 ± 0.038 mas, τ Ceti 2.022 ± 0.010 mas, 61 Cyg A 1.775 ± 0.013 mas). The dark-adapted naked eye sees a flash of 54–148 photons at the cornea (Hecht, Shlaer and Pirenne 1942), and a single photon slightly above chance (0.516 ± 0.010, Tinsley and others 2016). These data concern the reading, not the state of the objects.

What the author's framework had already said.

- The author's book NUMERI (v119, §§[01404]–[01422]) places the threshold in the ratio between the strength of the bond that holds an object together and its mass, a quantity with the dimensions of an acceleration, and notes its closeness in form to the galactic scale $a_0 = 1.2 \cdot 10^{-10}$ m/s², adding that this is a convergence of form, not a proven identity. Here it is reading T0.
- The status report of 26 August 2026 (version 2.9, file NON_SALDATO.md) had found that the threshold, used as a criterion, predicts nothing beyond standard decoherence, and that a version of it as a rate of collapse is excluded by the caesium atom of E8; it proposed to read it as a threshold on the reading, not on systems. Here it is reading T4.
- The «two walls» invariant of the status reports 2.9–2.18: the Compton wavelength of a mass and its Schwarzschild radius cross at $m = m_P/\sqrt{2} = 15.4$ μg. Here it is reading T1b.
- Deposit 2.38 counts the «layers» between the Planck scale and a mass m as $\ln(m_P/m)$: 44.0 for the proton, 51.5 for the electron (the electron value is in 2.38). Here it is readings T1 and T1c.
- Deposit 2.40 reads the time of a clock as the geometric mean of its two light sides. Here it is reading T2.
- Deposit 2.48: in the framework, probability passes through the Born rule, whose exponent 2 is taken from quantum mechanics, not derived.
- For the second part. Deposit 2.36.2: «distance is the measure of the fragmentation of a register», and the fragmentation behaves as a Brownian motion that deposits 1/r (the source of the potential). Deposit 2.62: a reading is a projection «in fractions» on the reader's own ruler, two readers share the common fractions, and the object they share appears within a few tens of copies; with an object of many values the common part grows in steps (working notes of 25 October 2026). Deposit 2.50 tested one translation of «coming into being and diffraction distance» for dark energy (none of 18 combinations within 0.02 of Ω_Λ). The phrase «fractional distance» has no rule in the earlier material.

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2. The thesis, motivated and described

2.1 The framework, in a few lines

In the author's framework the physical world is made of relations. Space and time share only one logical element, which the author calls their greatest common divisor (GCD), as the numbers 3 and 5 share only 1. The local world arises when

a relation is counted through this GCD: read by an observer inside it, the multiples appear as time and the fractions as space. Observed objects are related to the observer through a «diffraction», in fractions, of the observer's counting.

2.2 What the thesis says about quantum mechanics

The summary «MM in brief» published on every record says: when an observed object is large enough, its mass fixes it in the counting of the point of observation, and we see it in one place and at one instant. When it falls below a certain mass level, the principal GCD, that of the point of observation, and the GCD of the diffraction that generates the object become equal, as in a mirror: the object no longer occupies a single point but spreads over several possible positions, and we observe it as probability. The summary adds that the idea of a passage from quantum to classical behaviour that depends on mass is close to that of Penrose and Diósi.

Three things in this text are readings by the AI, approved by the author for publication: «principal GCD» = the GCD of the point of observation; «GCD of the diffraction» = that of the diffraction that generates the object; the threshold = the point where the two become equal. The author's own words are in AUTHOR_BACKGROUND.md.

Two later remarks of the author (25 October 2026), in the words of physics. First: the mass limit cannot be determined, because a calculation of the «fractional distance» is missing; at an adequate distance even a macroscopic object could take on quantum features. Second, in answer to the objection that the Moon is seen sharp from 384 000 km: distant bodies such as the Moon are large enough to stay fixed despite a high fractional distance; a direct test would be to look at a single stone on Mars with the naked eye, without technological mediation. The author calls the first a suspicion to be evaluated.

2.3 What has to be decided to give the threshold a number

- **What «quantum behaviour» means, as something measured.** We fixed two observables: stripes of interference of the centre of mass that differ from the classical prediction (Q1), or a state of motion that no classical description allows between distinguishable positions, shown by a negative Wigner function (Q2).
- **A mass alone is not enough.** Collapse models depend also on the separation Δx between the two positions and on the time T for which the superposition is kept. A threshold must say $m^*(\Delta x, T)$.
- **It must be intrinsic.** A threshold of the framework should be there even without an environment; where decoherence already destroys the stripes, a threshold cannot be seen, and this must be said.
- **What the two GCDs are, in physical quantities.** The framework does not say it. Each translation of §3 makes a choice; the choices are the AI's, and an outcome hits that choice, not the author's premise.
- **What «distance» is, and for whom.** Two things must be kept apart: the state of the object (is the object itself in a superposition?) and the reading (how does it appear to an observer who looks at it from far away?). Data on the Moon, on stars and on the eye concern the reading, never the state. Each reading of §3.3 says which one it speaks about.

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3. How the thesis was tested

3.1 The inventory (Level 0)

Twenty papers downloaded from arXiv (fingerprints in 00_fonti/IMPRONTE.txt; the papers are not redistributed). For each experiment: mass, separation, time, environment, outcome and the page where they are written. Before any calculation the experiments were divided into **decisive** (E1, E3, E8), **supporting**, which cannot exclude a reading on their own (E2 and E9, read second hand; E5), and **not decisive**, which show no superposition beyond the zero-point motion (E4, E6, E7). For E3 the effective mass is a choice of the authors: E3 decides a reading only if the outcome stays the same when the effective mass changes by a factor of 100 up or down.

3.2 The ten readings (Level 1)

For every pair (reading, experiment) a program computes a ratio r: $r < 1$ means the quantum side, $r > 1$ the side where the reading says «fixed in one place». The outcome rules were fixed before the calculation:

- **excluded** (registered label; in this text «does not follow from the data»): a decisive experiment shows quantum behaviour with $r > 10$;
- **compatible, put to the test:** no exclusion, and a decisive experiment has $0.1 \le r \le 10$;
- **not testable today:** no exclusion, and all decisive experiments have $r < 0.1$.

The factor 10 covers conventions of order one (2π between h and ħ, a factor 2 between the conventions of Diósi and of Penrose, Δx taken equal to the grating period).

reading	principal GCD read as	GCD of the diffraction read as	threshold	origin
T0	— (the book's criterion)	—	internal acceleration (bond / mass) below a_0	book NUMERI
T1	the GCD at the origin (Planck scale)	the GCD of the object	m* = m_P = 22 μg (layers ln(m_P/m) = 0)	AI, on 2.38
T1b	— (the two walls: the Compton wavelength of the object equals its Schwarzschild radius)	—	m* = m_P/√2 = 15.4 μg	book NUMERI; status reports 2.9–2.18
T1c	the GCD of our own scale (proton; variant: electron)	the GCD of the object	m* = proton mass (or electron mass)	AI, on 2.38
T1d	the mass of the observer	the mass of the object	m* = 70 kg	book NUMERI
T2	the light side of the observation, cT	the diffraction side of the object, h/(mc)	the geometric mean √(hT/m) equals Δx: m* = hT/Δx²	AI, on 2.40
T3a	the energy resolution of the observation, ħ/T	the gravitational energy E_G of the two positions, mass spread over the whole object	E_G·T = ħ	AI, on Penrose
T3b	as T3a	as T3a, nuclei spread over $R_0 = 2.54 \cdot 10^{-10}$ m	E_G·T = ħ (Diósi–Penrose at the experimental limit)	AI, on Diósi–Penrose
T4	the environment that records the position	—	no intrinsic threshold: decoherence rate × T = 1	status report 2.9
T5	the background, a_0	the acceleration G m/Δx² with which the object reads its own mass	G m/Δx² = a_0	AI, on the book

For T3a and T3b the gravitational energy of a uniform sphere displaced by Δx is that of Howl, Penrose and Fuentes (2019, eq. 23). For T4 the collision rate with the residual gas is that of Romero-Isart (2011, eq. 27). Every constant is written in the registered file (02_livello1/TRADUZIONI_LIVELLO1_v1.md, registered 22 October 2026 at 12:39:50 UTC, in Italian); there are no free parameters.

What was known before writing the readings. The registered predictions were written after rough estimates made by the AI from memory, and they say so: they are contaminated. All ten outcomes agreed with them. One numerical closeness was known before the calculation: m_P/√2 = 15.4 μg against the 16.2 μg of E3. By the rule on the effective mass it has no weight: it was registered as a coincidence without structure.

3.3 The second part: readings of the fractional distance

The same three steps, in the folder 09_distanza_frazionale: the inventory of the book, of the earlier deposits and of the experiments (Level 0, §1); eight readings registered with their outcome rules on 25 October 2026 at 12:59:58 UTC, before any calculation (02_livello1/TRADUZIONI_L1_v1.md, in Italian); the tests (Level 2, §5.5). Two observables: Q, the state of the object (as in the first part), and L, the reading of an observer (whether the object appears fixed within its own size, or spread over the diffraction figure of the observer's aperture, where each photon lands at random). The outcome rules are those of §3.2, with one addition: a quantity that a reading says is constant must not change by more than a factor 2 (or by more than 5σ) in a decisive experiment.

reading	the fractional distance is	how mass enters	observable
D0	the physical distance, with no rule of its own (standard quantum mechanics)	as in the first part	Q
D1	the separation of the two positions, measured in the scale at which a copy carried away by the environment tells them apart (for light, its wavelength λ: u = Δx/λ); after N copies the coherence is s(u)^N, and the object is fixed when the copies make its position common to the records of the environment (the threshold of deposit 2.62, tolerance 10^{-6})	through the number of copies per unit time (size,	Q

		temperature, pressure)	
D2a	the distance between the object and whoever records it, alone; the copies that nobody collects do not count	a mass limit growing with that distance	Q
D2b	the distance from the nearest record of any kind; it reduces to D1 (object not isolated) or D4 (isolated)	—	Q
D4	an informational distance: the fraction f of the copies of an outcome that reach the observer (f = 0 for a closed laboratory, as for Wigner's friend)	a macroscopic friend makes many copies: f = 0 needs total isolation	Q, for the observer
D5a, D5b	the fragmentation of deposit 2.36.2: its copies carry the image (D5a), or its deposited share $w = Gm/(2Lc^2)$ fixes the reading when $w \geq w^*$ (D5b, with $w^* = 1, 1/2, 1/(16\pi)$)	through the number of copies, or through w	L
D6	the diffraction cell of the copies (the light) in the observer's aperture a, at the distance L of the object, in units of the object's size D: $F = 1.22\,\lambda L/(aD)$; fixed if $F \leq 1$; for $F > 1$ the reading is the figure of the aperture, the same for every object smaller than the cell	only through the size (D6m: a mass limit growing with L)	L
D6b, D6c	with N copies collected the centre of the figure is known to $\delta/\sqrt{N}$ (D6b); below a threshold of copies there is no reading at all (D6c; for the eye 54–148 photons at the cornea)	through the number of copies	L
D7	the fractionation grows with the travel time of light from the observer, by $1 + \kappa\, L/(c\, t_0)$ with $\kappa = 4.05$ (a law tested on galaxy clusters in the working notes)	the mass limit shifts by the same factor	Q

The terms «diffraction cell», «parity» and «becoming equal» are bridges chosen by the AI: in D6 the summary's «the two GCDs become equal» is read as the reading taking on the grain of the observer ($F > 1$: the figure is that of the aperture, not of the object), and the second remark's «large enough to equalise the two GCDs» as the object's size being a multiple of that grain ($F \leq 1$). D4 and D6 are ideas of the AI; D1 gives the reading «the threshold is decoherence» (T4) a rule.

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4. What comes out

4.1 All outcomes

reading	decisive number	registered outcome	in words
T0	E1: sodium bound at $1.26\cdot10^{16}$ m/s², $r = 1.0\cdot10^{26}$; E8: $r = 2.2\cdot10^{31}$	excluded	does not follow from the data
T1c	E8: caesium, r = 132; E1: $r = 1.7\cdot10^{5}$ (electron variant: $2.4\cdot10^{5}$ and $3.1\cdot10^{8}$)	excluded	does not follow from the data
T5	E3: $r = 2.0\cdot10^{27}$, and at least $2\cdot10^{23}$ changing the effective mass by 100	excluded	does not follow from the data
T1	decisive experiments at most $r = 1.3\cdot10^{-14}$ (E1); E3 does not decide (0.0074–74)	not testable today	alive, not testable
T1b	as T1 (E3: 0.011–105)	not testable today	alive, not testable
T1d	E3: $2.3\cdot10^{-10}$	not testable today	alive, not testable
T3a	$r \leq 4.2\cdot10^{-10}$ (E8)	not testable today	alive, not testable
T3b	$r \leq 9.3\cdot10^{-14}$ (E1)	not testable today	alive, not testable
T2	E1: r = 1.24	compatible, put to the test	holds as a condition of grating interferometers, not as a general threshold (§4.3)
T4	E1: r = 0.29	compatible, put to the test	coincides with standard quantum mechanics

Programs and numbers: 02_livello1/calcoli_livello1_v3.py, results in risultati_livello1.json and log_livello1.txt.

4.2 The readings that do not follow from the data

- **T0, the internal acceleration of the book.** In E1 the sodium atoms of each cluster are held together by accelerations of about $1.3 \cdot 10^{16}$ m/s², 26 orders of magnitude above a_0, and yet the clusters interfere. The same holds for the caesium atom of E8. If the threshold were the internal acceleration compared with a_0, these objects should be fixed in one place.
- **T1c, the threshold at our own scale.** A caesium atom (132 proton masses) and clusters of 172 000 u behave quantum mechanically: the threshold is not at the mass of the proton, nor at that of the electron.
- **T5, a_0 as a mass.** For the vibrating crystal of E3 the reading would give a ratio of $2 \cdot 10^{27}$, and changing the effective mass by a factor of 100 does not save it. The link that the book suggests between galaxies and the quantum world does not carry over in this form.

What falls is each translation of the two GCDs into a physical quantity, which is a choice of the AI, not the author's premise.

4.3 The reading with the clock of deposit 2.40 (T2)

The reading. In deposit 2.40 the time of a clock is the geometric mean of its two light sides. Here the two sides are the diffraction side of the object, h/(mc), and the light side of the observation, cT. The space that the observed point occupies in the time T is then $\sqrt{(hT/m)}$, and the threshold is where this space equals the separation Δx: $m^* = hT/\Delta x^2$.

What the data say. In E1, with Δx equal to the grating period and T the flight time between two gratings, r = 1.24: the record experiment sits exactly at this threshold. With ħ in place of h the ratio would be 7.8, and it would exceed 10 as soon as the effective separation were 13 % larger than the period: the data prefer h. With h the outcome stays the same for separations up to about 2.5 times the period (02_livello1/prova_robustezza.py).

Why this is not a general threshold. The condition $m = hT/\Delta x^2$ is exactly the Talbot condition on which an interferometer with three gratings is designed: the first grating localises the object at the scale of the period, and the stripes form only if the wave spreads at least that much. The idea of T2 was written after reading this in the E1 paper, and it is declared as contamination. In quantum mechanics the coherence of a free object grows as $\hbar T/(m\sigma_0)$, where σ_0 is how localised it was at the start; T2 is the case $\sigma_0 \approx \Delta x$, up to numerical factors. Where the superposition is created by laser pulses, quantum mechanics goes far beyond T2: in the atom interferometer of Kovachy and others, with 54 cm and at most 3 s, $r \geq 2.1 \cdot 10^7$ (an experiment read second hand). In E8, during the phase in which the 1.9 μm separation is built (0.267 ms), r = 4.5. (02_livello1/t2_oltre_talbot.py; 02_livello1/CORREZIONE_T2_LIVELLO1.md.)

Outcome. With the registered rules T2 remains «compatible, put to the test». The reading «$m^* = hT/\Delta x^2$ is the threshold of quantum behaviour» does not follow: what remains is a narrower reading, in which T2 describes the condition of grating interferometers, that is, a threshold on the reading, not on the object. T2 is not identical to quantum mechanics: it is a particular case of it, which quantum mechanics exceeds.

4.4 Decoherence (T4)

In E1 the collisions with the residual gas remove about a quarter of the coherence if the gas is air at the stated pressure (r = 0.29), and about two thirds if it is hydrogen, the usual residual gas of an ultra-high vacuum (r = 1.08). This is the reading «threshold on the reading, not on systems» proposed in August. It coincides with standard quantum mechanics by construction. The second part gives it a rule, D1 (§4.6): the separation measured in the scale of the copies, and the number of copies.

4.5 The readings that cannot be tested today

- **T1, T1b, T1d: thresholds at the Planck scale or above.** No experiment has put an object heavier than about $3 \cdot 10^{-22}$ kg into a superposition as large as itself. The crystal of E3 (16.2 μg effective) does not decide, because its effective mass depends on a choice of the authors; the closeness of 15.4 μg (T1b) and 16.2 μg remains a coincidence without structure.
- **T3a, T3b: the readings of Diósi–Penrose type.** In today's experiments $E_G \cdot T/\hbar$ is at most $4.2 \cdot 10^{-10}$. A cross-check: for the crystal of E3 the formula of T3b gives the boundary at $R_0 \approx 2\text{–}4 \cdot 10^{-17}$ m, the same order of magnitude as a re-analysis of that experiment (Fadel 2023, pointed out by the verifier and not read in full).

An illustration (outside the protocol, not a test). For a silica object that must spread as much as its own size in free flight (02_livello1/illustrazioni.py): after 1 ms the T2 condition is met up to $3.3\cdot10^{-21}$ kg ($2\cdot10^{6}$ u, 14 nm), the T3a effect becomes of order one at $6.6\cdot10^{-14}$ kg (3.9 μm); after 100 s the two values are $3.3\cdot10^{-18}$ kg (0.14 μm) and $6.6\cdot10^{-17}$ kg (0.39 μm). A test of a gravitational threshold therefore needs long times and very localised initial states, or a spreading faster than free flight, or space.

4.6 The second part: distance

Programs: 09_distanza_frazionale/02_livello1/calcoli_L1.py (registered 25 October 2026, 13:01:40 UTC) and calcoli_L1_v2.py (13:26:57 UTC, after the first verification); results in the JSON files and logs next to them. All outcomes agreed with the predictions registered with the readings, which were contaminated: the AI knew the main papers before writing them.

reading	decisive number	outcome		
D0	correlations beyond the Bell bound at every separation tried, but no experiment changes only the distance from whoever records	not testable today (compatible)		
D1	one copy erases the contrast at $\Delta x = \lambda/2$ ($r = 1.00$); heated fullerenes and residual gas follow decoherence theory	compatible, put to the test (standard decoherence)		
D2a	same apparatus, same observer, only the molecules heated: visibility 47 % → 7 % (factor 6.7) → 0	does not follow from the data		
D2b	—	no outcome of its own (it reduces to D1 and D4)		
D4	friends coded in photonic paths or a few qubits: violations of the Bell-type and local-friendliness inequalities	compatible; not testable today as a rule (no datum on the leak of copies)		
D5a	predicted width of the image about 2 rad, against stars read as discs of $4\cdot10^{-8}$ rad ($r = 4.9\cdot10^{7}$)	does not follow, with the bridge «the copies of 2.36.2 carry the image»		
D5b	the Moon, read sharp, has $w = 7.1\cdot10^{-14}$; with $w^* = 1$, 1/2, $1/(16\pi)$ it would not be fixed	does not follow for these w^*; with w^* free, not testable today (only $w^* \le 7.1\cdot10^{-14}$)		
D6 (sharp boundary at $F = 1$)	HD 177724, a star with $F \approx 1.8$ at the longest baseline of CHARA in the K′ band, has a diameter measured to 2 % (0.897 ± 0.017 mas; found after the calculation)	does not follow from the data		
D6g (graded form, written after the calculation)	the reading depends on F only, continuously: for a disc the fringe visibility is \	$2J_1(x)/x$\	with $x = 3.83/F$; the Moon ($F = 0.01$ for the naked eye), α Cen A, τ Ceti and 61 Cyg A ($F = 0.19$–0.92 on the scale of CHARA)	compatible; it is wave optics, added after the calculation, not a blind test
D6m	Sirius B (1.018 $M_\odot$, 2.64 pc) too small by a factor 57 for CHARA in K′; τ Ceti (0.783 $M_\odot$, 3.65 pc) and 61 Cyg A resolved	does not follow, with the instrumental classification (decided in §5.5, block R)		
D6b	positions of unresolved stars known to 1/1500 of the cell of Hipparcos; the number of copies is required by the model, not counted	compatible; not testable today		

D6c	a stone of 10 cm on Mars seen by the naked eye at the closest distance: 3–6·10⁻⁹ photons per second at the cornea, one every 5–10 years, against 54–148 in 0.1 s	compatible: no reliable visual reading
D7	the correction is 3·10⁻²⁶ in a laboratory, 1.7·10⁻¹⁵ at Mars, 1.3·10⁻⁹ at α Cen	not testable today

What falls. The reading «for the state of an object only the distance from the observer counts»: changing only the environment, at the same apparatus and the same observer, changes the stripes. The bridge «the copies of the fragmentation of deposit 2.36.2 carry the image», and the three values of w* tried for its deposited share (with w* free, D5b remains not testable today); the deposit itself, which with those copies reads the potential 1/r, is not touched. A mass-only limit of the reading growing with distance, in the sense of block R (§5.5): with the registered instrumental classification the sample does not support it. And the sharp form of the cell: below it the size can still be read, gradually.

What holds, and what it says. For the state of the object (D1) the fractional distance is the separation measured in the scale at which the copies of the environment tell the two positions apart; the boundary is reached when the copies make the position common to the records of the environment: $N(m, T)\cdot(-2 \ln s(u)) = \ln(1/\delta)$. About 40 copies are needed at a sixth of the scale, 10 000 at a hundredth, about $2\cdot10^{34}$ for the crystal of E3 with the thermal photons of a cryostat at 10 mK. For the reading (D6g) the fractional distance is the observer's cell at the distance of the object, in units of the object, and the reading depends on it continuously; the many copies of a bright body fix its position inside the cell, as the second remark of the author says. In both, mass enters only through the number of copies and the size: in these two readings there is no single mass limit, and two objects of the same mass can fall on opposite sides (two stars; a hot and a cold molecule). The data test decoherence and optics; that the boundary is the point where the object becomes the common part of the records is the framework's reading.

The stone on Mars. Seen by the naked eye it sends too few copies for any reading (one photon every 5–10 years against a threshold of 54–148 in a tenth of a second); a rock would have to be 30–40 km across to reach the threshold, and it would still be mixed with the light of Mars, which sends $4.6\cdot10^{15}$ times more. The test cannot be done with the eye, and not for a limit of technology: the copies are missing. That for such an observer the stone «casts no shadow in the record» (deposit 2.62) is the framework's reading, not a result of the count. For the state of the object the distance that counts is not the distance from the observer but the informational one: that no copy of the position leaves towards anyone.

A proposal, not a result. D1 and D6g have the same form, a size to be told apart against the cell of the copies of whoever reads, with the number of copies; they could be one rule of the framework, close to the author's earlier idea that the fractionation of an object is read from its distance of diffraction from the observer (AUTHOR_BACKGROUND.md). The data do not say so: the quantities are different.

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5. The dated prediction (Level 2)

What was registered, and when. The prediction (05_livello2/PREVISIONE_LIVELLO2_v1.md) and its program were registered on 22 October 2026 at 13:00:30 UTC; that time is the time limit of the targets. Just before, a search of the arXiv metadata (June–October 2026) found no experiment in the informative region. It covers T1, T1b, T3a and T3b. Two corrections were registered a few minutes later, when no target could yet exist: the first on the parameters of MAQRO (§5.3), the second after the verification (§8). The rules valid today are those of CORREZIONE_2_PREVISIONE.md and of the program valuta_previsione_v2.py: if text and program differ, the program holds.

5.1 Which papers count

A paper counts if its first arXiv version (or the journal, if earlier) is dated after the time limit and no later than 31 December 2035, if it reports an experiment (not a proposal), and if it gives the mass, the separation, the time, the size and material when needed, and the observable (Q1 or Q2) with its significance. Confidence levels and p-values are converted into standard deviations with fixed formulas. The target is the first paper, in order of date, with a ratio of at least 0.1.

5.2 The rules

- **Planck block (T1, T1b), r = m/m*.** Prediction: above m* no quantum behaviour of the centre of mass. With quantum behaviour seen at 3σ or more: «compatible, put to the test» for $0.1 \le r < 1$, «tension» for $1 \le r \le 10$, «excluded» for $r > 10$; quantum behaviour seen at 3σ or more with $r > 10$ gives «excluded» whatever the visibility. «Favourable» only if quantum behaviour is not seen at 3σ or more, $r \ge 1$, the stripes vanish above m*

(fraction of the expected visibility $f + 2\sigma_f < 0.2$), and in the same apparatus, with a mass of at most $m^*/2$, they are there ($f - 2\sigma_f > 0.5$), with no known cause given by the authors. For a vibration mode, the outcome must hold for every effective mass within a factor of 100.

- **Gravity block (T3a, T3b), $r = E_G \cdot T/\hbar$.** Prediction: the superposition is reduced by a factor $\exp(-r)$ beyond the known decoherence. With quantum behaviour seen at 3σ or more: «compatible» for $0.1 \le r \le 1$, «tension» for $1 < r \le 10$, «excluded» for $r > 10$; «excluded» also if the observed fraction of visibility is clearly above what the reading allows ($f - 2\sigma_f > 10 \cdot \exp(-r/2)$). «Favourable» if $r \ge 1$, the observed fraction of visibility is clearly below one ($f + 2\sigma_f < 0.8$) and lies in the band allowed by the reading with a factor 2 on r ($\exp(-2r)/3 \le f \le 3 \cdot \exp(-r/2)$), and the authors give no known cause for the loss. In this block the rules on f come before those on σ, because T3 predicts a reduced interference, not an absent one.
- The binding definition of all these rules is the program valuta_previsione_v2.py.
- If no paper reaches $r \ge 0.1$ by 31 December 2035, the block is «not assessable».
- A favourable outcome for T3 would not distinguish the author's framework from Penrose or from Diósi–Penrose: the formulas are theirs. It is written so in the prediction.

5.3 What the planned experiments would reach

The valid table is 05_livello2/log_progetti_v2.txt (the tables printed in log_previsione.txt and log_previsione_v2.txt use the earlier, wrong parameters of MAQRO; see correction 1). With the parameters of the MAQRO study (grating period 50 nm, silica at 1 850 kg/m³):

project	mass	r for T3a (Δx = 50 nm / 100 nm)	r for T1b
MAQRO, 100 s (study scenario)	10^{11} u	0.095 / 0.35	$1.1 \cdot 10^{-8}$
MAQRO, 100 s (study scenario)	10^{10} u	0.0087 / 0.029	$1.1 \cdot 10^{-9}$
MAQRO within the feasibility study (400 nm, 40 s)	up to about $3.7 \cdot 10^{10}$ u	at most 0.05	—
E1, next step at 1 MDa and 25 m/s	10^6 u	$2.2 \cdot 10^{-11}$	$1.1 \cdot 10^{-13}$
E1, vertical interferometer	$1.7 \cdot 10^7$ u	$3.6 \cdot 10^{-8}$	$1.9 \cdot 10^{-12}$
the target of Carlesso and others for the rate of Ghirardi, Rimini and Weber	10^7 u	$3.3 \cdot 10^{-7}$	$1.1 \cdot 10^{-12}$

Only the most optimistic MAQRO scenario, with a separation of two grating periods, enters the region where the rules give an answer ($r \ge 0.1$), and only as «compatible, put to the test»: far from a clear outcome. Experiments with atoms or molecules below 10^7 u stay below 0.1 for both blocks even with T up to 1 000 s and Δx up to 1 m. Before the correction, the AI had estimated from memory a ratio of about 6 for MAQRO: with the right parameters it is 0.095–0.35, and the test written with that estimate («r between 1 and 10») does not hold. It is reported as it is.

5.4 What the prediction does not cover

- A new non-interferometric limit on Diósi–Penrose (X-rays, heating) concerns the radius R_0 of T3b, not T3a, which is a criterion without dynamics.
- T2 and T4: T4 has no prediction different from quantum mechanics. T2 would have one (in MAQRO it would differ from quantum mechanics), but atom interferometers already exceed it; a dated prediction for T2 was not made.

How to apply it. Import valuta_previsione_v2.py and call valuta_P() or valuta_G() with the numbers of the paper; the program applies the rules and the admission criteria. Its 45 tests (log_previsione_v2.txt) include the exact edges r = 0.1, 1 and 10 and the counter-examples raised by the verifier.

5.5 Tests of the distance readings (second part)

Registered on 25 October 2026 at 13:08:06 UTC (09_distanza_frazionale/03_livello2/PREVISIONE_L2_v1.md), before the two primary sources that decide block R were downloaded (13:25:09 UTC). The values that decide it were known from the abstracts, and the expected outcome was written: it is the application of a registered rule to primary sources read afterwards, not a blind test.

- **Block R, decided: is the limit of the reading a mass or a size?** A mass limit growing with distance would have to lie above 1.02 solar masses at 2.64 pc (Sirius B is not fixed in the reading: its predicted disc, 0.029 mas, is below a tenth of the finest cell of CHARA in K′, 0.163 mas) and below 0.78 at 3.65 pc (τ Ceti, 2.022 ± 0.010 mas, mass 0.783 ± 0.012 from asteroseismology with the interferometric radius); the same with 61 Cyg A (1.775 ± 0.013 mas, 3.49 pc, masses 0.67–0.74 from orbits, discordant). **With the registered instrumental classification, the sample does not support a mass-only limit growing with distance.** It is compatible with the reading by size, which it does not make necessary. Program valuta_L2_R.py (registered 13:27:55 UTC).
- **Block Q, the state of the object.** No new target: the test is the prediction of §5, in which the fraction f between the observed visibility and the one expected with known decoherence is exactly the comparison with D1. One rule is added: D1 «is not enough» if an admitted paper gives $f + 2\sigma_f < 0.5$ (more than half of the coherence lost, at 2σ) with a mass above 10^{-21} kg and no known cause, where a known cause is one given by the authors with an estimate that accounts for at least half of the missing visibility; σ_f is the total uncertainty given by the authors. (The rule was first registered with the wrong sign, $f - 2\sigma_f$; the verifier found it and it was corrected before any paper could count.)
- **Block W, a proposal: Wigner's friend with a massive record.** A local-friendliness test in which each friend's record is a massive mechanical mode, like the 16-microgram crystal of E3, would decide between D4 (no mass limit if no copy leaves) and the Planck-scale readings T1 and T1b of the first part. With the rule on the effective mass (a factor
- the records should have at least 2.18 mg (T1) or 1.54 mg (T1b) of effective mass. To become a protocol it still needs: a scheme with two laboratories, a way for the mode to record two outcomes and to undo it, the measurements of the outside observers, a model of the leak of copies, and the effect of the fidelities on the inequality. The cat state of E3 (fidelity about 66 %) is a step, not yet a record of an outcome. No experiment with this combination was found; related proposals and experiments exist (Wigner's-friend scenarios with macroscopic cat states; superpositions of massive resonators; the program of local-friendliness tests on quantum computers). Deadline 31 December 2035, like §5.

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6. What is not claimed

- No mass threshold is derived, and none has been found: no experiment has met one.
- It is not claimed that quantum mechanics is explained by the author's framework. Quantum mechanics predicts the probabilities by itself; in the framework the exponent 2 of the Born rule is taken from it (deposit 2.48).
- The compatibility of T2 and T4 with the data is coherence with facts that quantum mechanics already predicts, not an independent test of the framework.
- The readings that do not follow from the data hit the AI's translations of the two GCDs, not the author's premise. The readings that remain (Planck scale, mass of the observer, Diósi–Penrose type) are not confirmed: they cannot be tested today.
- The closeness of the Planck-scale reading to the crystal of E3 (15.4 against 16.2 μg) is not a result.
- The closeness to Penrose and Diósi written in the summary «MM in brief» is, after this work, a possibility still to be tested; a favourable outcome would not tell the framework apart from Diósi–Penrose.
- Second part. The two rules of the fractional distance that hold (D1 for the state, D6g for the reading) coincide with decoherence and with wave optics: the framework admits them, it does not predict them differently. D6g was written after the calculation. That the two rules are one rule of the framework is a proposal.
- The fringes of starlight are a fact of the reading, not of the state of the star: no star is claimed to be in a superposition. The non-resolution of Sirius B is an instrumental inference from its radius, not a measured disc.
- The count of photons from the stone on Mars says that the naked eye gets no reliable reading; it does not say that the stone does not exist.
- No experiment changes only the distance between a system and whoever records it, and no Wigner's friend has a macroscopic mass: those questions stay open.

7. How to reproduce

Seven programs in Python 3, with no external libraries, a few seconds in all: calcoli_livello1_v3.py (the registered calculation of the ten readings), illustrazioni.py, prova_robustezza.py and t2_oltre_talbot.py (the illustration and the checks after the calculation) in 02_livello1; valuta_previsione.py, progetti_v2.py and valuta_previsione_v2.py (the prediction, the table of the projects, the rules after correction 2) in 05_livello2. Every protocol and every program has a SHA-256 fingerprint and a UTC time registered before the calculation; the results are in JSON files and logs. run_all.py re-runs the seven programs and compares every number with the registered ones (§ «How to run»). Two earlier versions of the first program (calcoli_livello1.py and _v2.py) stopped at an internal check that was itself wrong, before producing any result (DEVIAZIONI_LIVELLO1.md and _D3.md); they are kept for the record. The arXiv search (controllo_arxiv.py) and the text extraction from the papers (estrai_testo.py) are not re-run: the first depends on the day it is run, the second needs the papers, which are not redistributed.

For the second part, three more programs in 09_distanza_frazionale, with no external libraries: calcoli_L1.py (the registered calculation of the eight readings), calcoli_L1_v2.py (the corrections after the first verification) and 03_livello2/valuta_L2_R.py (block R). run_all.py re-runs them too: ten programs in all. The two tools that extract and search the text of the sources (00_fonti_estrai_testo.py, 00_fonti_cerca.py) are not re-run: they need the papers.

8. Provenance

The author is not a physicist. The inventory, the translations, the programs and the calculations were made with an artificial intelligence (Claude, Anthropic) on 22 October 2026 and verified by a second one (gpt-6-luna via Codex, OpenAI), one round for each level.

- **Level 1 (six remarks).** Accepted: the second value of R_0 found in two reviews ($4.94 \cdot 10^{-10}$ m) is not a limit on Diósi–Penrose but a CSL limit of the same paper, copied wrongly; the error bars of E5 are 95 % intervals, not standard deviations; E2 cannot exclude by itself; the time of E3 is an asymptotic estimate. Accepted in part: the sensitivity of T2 to the separation and to h or ħ, now declared (§4.3). No outcome changed.
- **Level 2 (six remarks).** Accepted: the program did not yet apply the admission rules (now it does); the margin of T3 was ambiguous (now the thresholds apply to r with no margin, the factor 2 only in the rules on visibility); in the Planck block «favourable» could prevail over «excluded» (now it cannot); the MAQRO rows at 100 s are study scenarios, not performances; T2 has the scale of free spreading only up to a factor 2π.
- **Errors found by the AI itself:** two internal checks written wrongly (corrected before any result); the meaning of T2, first presented as «coinciding with quantum mechanics» (corrected: it is a particular case, exceeded by atom interferometers); the parameters of MAQRO (corrected before any target could exist).
- Both rounds of the verifier found the word «confirmation» in the summary «MM in brief» stronger than what these data show; the summary is the author's text and was not changed here (§6).
- **The package** was then verified by the same verifier, at most in two rounds. The first round (ten remarks) corrected the summary of the rules in §5.2, the count of the readings that cannot be tested today (five, not two), a reference, a note that clarifies the translation of one sentence of the author, and the list of registers in the archive. The second and last round found these corrections right and led to a fuller list of contents in the Italian text and to this sentence. The reports and the answers are in VERIFICA_PACCHETTO.
- **The second part** (25 October 2026) was done with the same tools and verified by the same verifier in three runs: two rounds for Level 0, two for Level 1, one for Level 2. Every round said «not ready» and every correction was applied; the arithmetic was confirmed each time. The main ones: the experiments on distance are not a controlled series, so «no trend with distance» was withdrawn and D0 became «not testable today»; D4 and D6b were downgraded to «not testable today», because the leak of copies and the number of copies are not measured; the sharp form of D6 does not follow from the data (HD 177724), and the graded form is declared as written after the calculation; the conclusion of block R was narrowed; the rule of block Q had the wrong sign; block W became a proposal. The reports are in 09_distanza_frazionale/04_verifica_astra, the answers in the outcome notes (Italian). The extended package was then verified again (round 3 on the package: five remarks, all accepted; the main result is now stated for the two readings of distance only, since the Planck-scale readings of the first part remain untested; block R is stated with its restriction in every summary; D5a and D5b are kept apart; the description says that block R was not a blind test and that the non-resolution of Sirius B is inferred). Round 4, the last, found no remark and called the package ready. Reports and answers in VERIFICA_PACCHETTO.

The author's sentences from which the work arises are in AUTHOR_BACKGROUND.md.

Glossary

- **a_0:** the acceleration scale of galaxies, about $1.2 \cdot 10^{-10}$ m/s², below which their rotation departs from Newton's law.
- **Atomic mass unit (u):** $1.66 \cdot 10^{-27}$ kg, about the mass of a proton; 1 kDa = 1 000 u, 1 MDa = 10^6 u.
- **Cat state:** a superposition of two clearly different states of a large system.
- **Centre of mass:** the point that moves as if all the mass were there; a superposition of the centre of mass means the whole object is in two places.
- **Collapse model:** a modification of quantum mechanics that makes superpositions end by themselves, faster for larger objects (GRW, CSL, Diósi–Penrose).
- **Compton wavelength:** h/(mc), the length associated with a mass m; the reduced one is ħ/(mc).
- **Contamination:** what was known of the result when a rule was written.
- **Copy:** in the author's framework, each photon or molecule that carries away a trace of where an object is.
- **CSL:** continuous spontaneous localisation, the continuous version of the GRW model, with a rate λ and a length r_C.

- **Decoherence:** the loss of the stripes caused by the environment, which records where the object is.
- **Decisive, supporting, not decisive:** the classes into which the experiments were put before the calculation.
- **Diffraction cell:** the smallest angle an aperture of size a can resolve, about 1.22 λ/a; multiplied by the distance, the smallest size it can resolve there.
- **Diósi–Penrose model:** a superposition of two positions of a mass lasts about ħ/E_G; with a radius R_0 for the nuclei.
- **Effective mass:** the mass assigned to a vibration mode; it depends on how the authors define its amplitude.
- **E_G:** the gravitational energy of the difference between the two arrangements of mass of a superposition.
- **Fractional distance:** a term of the author; in this work, the size to be told apart measured in the cell of the copies of whoever reads (§3.3).
- **GCD:** greatest common divisor; in the author's framework, the only logical element that space and time share. «Principal GCD» and «GCD of the diffraction» are terms of the framework (§2.2).
- **GRW:** the collapse model of Ghirardi, Rimini and Weber (1986).
- **Interferometer:** an apparatus that splits a wave and recombines it, so that the parts can interfere; a Talbot–Lau interferometer uses three gratings. A stellar interferometer combines the light of a star collected by telescopes a distance B apart (the baseline).
- **Local friendliness:** inequalities that hold if the outcomes observed by «friends» inside closed laboratories are absolute, the agents are free and influences are local; quantum mechanics violates them (Bong and others 2020).
- **Macroscopicity μ:** a number that says how strongly an experiment restricts collapse-type modifications of quantum mechanics (Nimmrichter and Hornberger 2013).
- **MAQRO:** a proposed space mission for interference with particles of 10^7–10^{11} u.
- **mas:** milliarcsecond, $4.85 \cdot 10^{-9}$ rad.
- **Planck mass (m_P):** $\sqrt{(\hbar c/G)} = 2.18 \cdot 10^{-8}$ kg, about 22 micrograms.
- **Ratio r:** for each reading and experiment, how far the experiment is from the threshold of the reading: r < 1 on the quantum side.
- **Reading (L) and state (Q):** how an object appears in the record of an observer, and what the object itself is doing; the second part keeps the two apart.
- **Registered:** written in a file whose SHA-256 fingerprint and UTC time were recorded before the calculation; the file is not changed afterwards.
- **Resolved:** an object whose angular size an instrument can measure; unresolved objects appear as the instrument's diffraction figure.
- **Schwarzschild radius:** $2Gm/c^2$, the radius at which a mass m would become a black hole.
- **Standard deviation (σ):** the measure of statistical uncertainty; 3σ corresponds to a confidence of 99.73 %.
- **Superposition:** the state of an object spread over several positions (or states) at the same time.
- **Talbot condition:** the condition under which a three-grating interferometer forms stripes: the wave must spread at least by the grating period between two gratings.
- **van Cittert–Zernike theorem:** the light of a distant, incoherent source of angular diameter θ is coherent over a transverse distance of about λ/θ; for a uniform disc the fringe visibility with baseline B is $|2J_1(x)/x|$, $x = \pi\theta B/\lambda$.
- **Visibility:** the contrast of the stripes, from 0 (none) to 1.
- **Wigner function:** a description of a quantum state in position and momentum; where it is negative, no classical description is possible.
- **Wigner's friend:** a thought experiment, now realised with photons and qubits, in which an observer (the friend) measures inside a closed laboratory that an outside observer treats as a quantum system.

Package contents

- README.md (this text); LEGGIMI_IT.md (the same text in Italian); AUTHOR_BACKGROUND.md (the author's sentences, kept apart from the main text); LICENSE.md.
- 07_soglia_massa/01_livello0: the inventory of the data (version 1, values read in the papers, with page numbers), its corrections after the verification, the program that extracted the text from the papers.
- 07_soglia_massa/00_fonti: IMPRONTE.txt, the fingerprints of the 20 papers downloaded from arXiv (the papers are not redistributed; they are listed in the References).
- 07_soglia_massa/02_livello1: the ten readings with the outcome rules (registered before the calculation), the three versions of the program and the two notes on the deviations, the results (JSON and log), the checks after the verification, the program on T2 beyond the Talbot geometry, the correction on T2, the illustration, the three outcome notes (v3 is the final one).
- 07_soglia_massa/05_livello2: the arXiv search before the prediction, the dated prediction and its program, the two corrections with their programs and results, the final note.
- 07_soglia_massa/03_verifica_astra: the two verification reports on the calculations (gpt-6-luna via Codex) and the answers to them; VERIFICA_PACCHETTO: the reports of the same verifier on the package and the answers.
- 07_soglia_massa/04_riquadro: only REGISTRO.txt, with the fingerprints and times of the proposals made to the

author for the wording of the summary «MM in brief»; the proposals are not part of this deposit.
- REGISTRO.txt in every folder: SHA-256 fingerprint and UTC time of every registration. A few registered working files are not in the archive: the author's premise with his exact words, the provisional version of the inventory (with the list of downloads asked of the author), the messages sent to the verifier (which contain the whole material) and the programs that composed them, and the proposals made to the author for the wording of the summary «MM in brief»; their fingerprints are in the REGISTRO.txt files, and they can be shown on request.
- 09_distanza_frazionale (the second part; one REGISTRO.txt for the whole folder, with relative paths): the two first evaluations of the author's remarks (NOTA_VALUTAZIONE_v1 and v2, written before the protocol: contamination); 00_fonti, the fingerprints of the 22 sources downloaded with the author's permission (not redistributed), and the two tools that extract and search their text; 01_livello0, the inventory of the book and of the deposits, the inventory of the experiments, its correction and the outcome note; 02_livello1, the eight readings with the outcome rules (registered before the calculation), the two programs with results and logs, the three outcome notes (v3 is the final one); 03_livello2, the tests, the program of block R with its results, the two outcome notes (v2 is the final one); 04_verifica_astra, the three reports of the verifier.
- run_all.py: re-runs the calculations and compares them with the registered ones. IMPRONTE.sha256: fingerprints of all the other files in the archive.
- Also left out of the archive, for the second part: the author's two remarks with his exact words (they are in AUTHOR_BACKGROUND.md), the messages sent to the verifier and the programs that composed them, the report written for the author and its correction, the sources and their extracted text; their fingerprints are in the REGISTRO.txt of 09_distanza_frazionale.
- In the working records this deposit is called «soglia di massa» (folder MM_sviluppo_quantistica/07_soglia_massa) and, for the second part, «distanza frazionale» (folder 09_distanza_frazionale); file names and registered documents keep their Italian names and language.

How to run

python run_all.py [output_folder]

The script runs the ten programs into a new or empty folder (a temporary one if none is given) and compares their results with the registered ones: the JSON files number by number, the logs line by line (the text must be identical; the numbers are compared with a relative tolerance of 10^{-9} of the larger value, with no absolute floor; line endings and empty lines at the end of a log are not compared). It ends with «ALL MATCH». Tested with Python 3.13.13 (the version is written in the record of the test from scratch); no other library is needed; a few seconds.

Related records

- 2.64 (DOI 10.5281/zenodo.23610071): mass as a contained rhythm; it announced this work.
- 2.40 (DOI 10.5281/zenodo.23280168): the clock as the geometric mean of two light sides (reading T2).
- 2.38 (DOI 10.5281/zenodo.23055978): the layers ln(m_P/m) (readings T1, T1c).
- 2.48 (DOI 10.5281/zenodo.23352814): Born rule and Tsirelson bound; the exponent 2 is imported.
- Status reports 2.0–2.27 (concept DOI 10.5281/zenodo.21802475): the two-walls invariant (2.9–2.18) and the August note on the threshold (2.9).
- 2.62 (DOI 10.5281/zenodo.23609967): objects as the common part of two points of view; the ruler in fractions and the threshold of copies (readings D1, D4, D6c).
- 2.36.2 (DOI 10.5281/zenodo.23287965): distance as the fragmentation of a register (readings D5a, D5b).
- 2.50 (DOI 10.5281/zenodo.23376729): an earlier translation of «coming into being and diffraction distance».

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